

MD MOMINUL HAQUE

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Profile

Graduate AI researcher specializing in resource-efficient computer vision, explainable deep learning (XAI), and multi-modal image segmentation. Proven expertise in designing and optimizing hybrid CNN-Transformer architectures, Vision Transformers (ViTs), and continuous test-time adaptation (TTA) frameworks using PyTorch and TensorFlow. A published first-author researcher experienced in translating complex mathematical modeling and algorithm design into robust, explainable diagnostic systems for medical and geospatial imaging.

Education

Universiti Teknologi PETRONAS (UTP) (*QS 2026: 251 | THE: 201-250*) Perak, Malaysia
Master of Science in Information Technology (By Research) 2024 – June 2026

Funding: Fully funded under the UTP Graduate Assistantship (GA) Scheme

Thesis: *MVT-Net: A Lightweight and Explainable Hybrid CNN-Transformer Framework for Retinal Vessel Segmentation*

Stamford University Bangladesh (*Top 5% of Class*) Dhaka, Bangladesh
Bachelor of Science in Computer Science & Engineering CGPA: 3.80 / 4.00

Thesis: *Attention-based Long-term Recurrent Convolutional Network for Physiotherapy Exercise Classification*

Technical Skills & Research Competencies

Programming: Python, C, C++

DL/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn

Libraries: OpenCV, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Albumentations

Web & Databases: HTML, CSS, SQL

Languages: English (IELTS 6.5), Bengali

DL & CV Methodologies: Image & Semantic Segmentation, Object Detection, Explainable AI (XAI / Grad-CAM++), Transfer Learning, Test-Time Adaptation (TTA), Attention Mechanisms

Tools & Systems: Git/GitHub, Overleaf (LaTeX), VS Code, Jupyter, Linux, Windows, Cloud Platforms

Experience

Graduate Teaching Assistant Sep 2024 – June 2026
Universiti Teknologi PETRONAS Perak, Malaysia

AI Engineer Jul 2024 – Sep 2024
Eravend Innovation Private Ltd. Dhaka, Bangladesh

Research Assistant (Remote) Mar 2024 – Sep 2024
Canada Syntex Machine Learning Alberta, Canada

Computer Science Teacher Dec 2023 – Mar 2024
World School of Bangladesh Gazipur, Bangladesh

Artificial Intelligence (AI) Intern Aug 2023 – Nov 2023
Inflexionpoint Technologies BD Ltd. Dhaka, Bangladesh

Leadership & Awards

Executive Committee – International Affairs and Student Mobility 2025 – Present

Student Representative Council (SRC), Universiti Teknologi PETRONAS (UTP)

Elected student leader as ambassador for international students in UTP.

PETRONAS Focused Recognition Award May 2026

Centre for Excellence in Teaching and Learning (EdEx), Universiti Teknologi PETRONAS

Awarded for technical leadership in designing and piloting the UTP Staff Continuous Professional Development (CPD) Dashboard, recognized for driving core institutional cultural beliefs and impacting academic career progression tracking.

Best Presenter Awards

– 2025 IEEE 9th International Conference on Software Engineering & Computer Science (ICSECS 2025), Malaysia.

– 2024 IEEE 9th International Conference on Computational Intelligence and Applications (ICCIA 2024), China.

Manuscript Reviewer – Elsevier Biomedical Signal Processing and Control (Q1 Journal)

Reviewed scientific manuscripts related to deep learning and biomedical image analysis.

Publications

- [1] Md Mominul Haque, Saipunidzam Mahamad, Suziah Sulaiman, Abdullateef Oluwabemiga Balogun, and Hussaini Mamman. “ToxoSegFusion: Attention-enhanced Dual-backbone Neural Architecture for Retinal Lesion Segmentation”. In: *IIUM Engineering Journal* 27.1 (2026). (**SJR Quartile: Q3**), pp. 321–347. DOI: 10.31436/iiumej.v27i1.3806.
- [2] Md Mominul Haque, Saipunidzam Mahamad, Suziah Sulaiman, and Abdullateef Oluwabemiga Balogun. “Noise-Space Adaptive Multi-Domain Filtering: A Unified Framework for Low-Light Urban Surveillance and Medical Retinal Imaging”. In: *Applied Research* 5.3 (2026). (**SJR Quartile: Q2**), e70106. DOI: <https://doi.org/10.1002/app1.70106>.
- [3] Md Mominul Haque, Saipunidzam Mahamad, Suziah Sulaiman, Abdullateef Oluwabemiga Balogun, and Kanneh Momoh. “DAFNet-Res: A Hybrid-Attentive Decoder Network with CLAHE-Enhanced Supervision for Retinal Vessel Segmentation”. In: *2025 IEEE 9th International Conference on Software Engineering Computer Systems (ICSECS)*. 2025, pp. 326–331. DOI: 10.1109/ICSECS65227.2025.11279243.
- [4] Saipunidzam Mahamad, Yi Han Chin, Nur Izzah Nasuha Zulmuksah, Md Mominul Haque, Muhammad Shaheen, and Kanwal Nisar. “Technical Review: Architecting an AI-Driven Decision Support System for Enhanced Online Learning and Assessment”. In: *Future Internet* 17.9 (2025). (**SJR Quartile: Q1**). ISSN: 1999-5903. DOI: 10.3390/fi17090383.
- [5] Md. Mominul Haque, Sweetly Akter, and Adnan Ferdous Ashrafi. “SwinMedNet: Leveraging Swin Transformer for Robust Diabetic Retinopathy Classification from the RetinaMNIST2D Dataset”. In: *2024 6th International Conference on Electrical Engineering and Information Communication Technology (ICEEICT)*. 2024, pp. 1286–1291. DOI: 10.1109/ICEEICT62016.2024.10534544.
- [6] Md Mominul Haque, Mohammed Al-Mukhtar, MD Samiul Islam, Alhassan Alie Kamara, and Mehedi Hasan Nirob. “A Case Study of Human Segmentation in Multispectral Imaging”. In: *2024 IEEE 9th International Conference on Computational Intelligence and Applications (ICCIA)*. 2024, pp. 155–160. DOI: 10.1109/ICCIA62557.2024.10719107.

- [7] Manik Hossain, Sumaiya Binte Siraj, Md. Mominul Haque, Gihun Song, Byungyong Ryu, and Md. Tauhid Bin Iqbal. "Physiotherapy Exercise Classification by Exploring Knowledge Distillation with Attention Modules". In: *2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*. 2023, pp. 1–11. DOI: 10.1109/NCIM59001.2023.10212747.

Manuscripts Under Review

- **Md Mominul Haque**, Adnan Ferdous Ashrafi, Saipunidzam Mahamad, & Sweetey Akter. "Swin-MedNetV2: A Hierarchical Transformer for Cross-Modal Medical Image Classification," *Neural Computing and Applications*. **(SJR Quartile: Q1)**
- **Md Mominul Haque**, Hosna Asma Ull, Saipunidzam Mahamad, Suziah Sulaiman, & Nur Alam. "MVT-Net: An Efficient Hybrid CNN–Transformer for Robust Retinal Vessel Segmentation Across Heterogeneous Datasets," *Biomedical Signal Processing and Control*. **(SJR Quartile: Q1)**
- **Md Mominul Haque**, Hosna Asma Ull, & Nur Alam MD. "Null-Space Gradient Projection for Forgetting-Free Test-Time Adaptation in Echocardiographic Segmentation," *IEEE Journal of Biomedical and Health Informatics*. **(SJR Quartile: Q1)**
- **Md Mominul Haque**, Saipunidzam Mahamad, MD Samiul Islam, & Fakhreldin Ali. "Lightweight MobileNetV2 U-Net with SCSE Attention for Efficient Multi-Class Land Cover Classification on the DeepGlobe Dataset," *International Conference on Sustainable Living (ICSL 2026 / ESTCON 2026)*.

Projects

CPD Learning Hours Dashboard for UTP Staff - A live dashboard developed for Universiti Teknologi PETRONAS (UTP). The system provides real-time analytics and reporting on training hours, compliance tracking, and professional development progress.

Face Recognition – Developed real-time facial recognition for EraVend GmbH.

Real-Time Face Mask Detection – Implemented MobileNetV2 for real-time mask detection.

Object Detection (Custom Data, YOLO API) – Built domain-specific YOLO object detector.

Facial Expression Recognition – Trained CNN model using PyTorch and custom datasets.

Bangla Handwritten Digit Recognition – Implemented Inception-GoogLeNet on Ekush dataset.

Online Crime Reporting System – Web-based platform to improve public crime reporting.

Departmental Shop Management – Inventory and sales management system.

Courses & Certifications

IBM AI Engineering Professional Certificate (In Progress) – Coursera

Deep Learning Specialization – Coursera

DeepLearning.AI TensorFlow Developer – Coursera

Generative Adversarial Networks (GANs) Specialization – Coursera

Neural Networks and Deep Learning – Coursera

Computer Vision Basics – Coursera

Cloud Computing Basics (Cloud 101) – Coursera